

limit of a product need not equal the product of the direct limits. For finite wedge sums there is no difficulty, so we do have a cohomology theory for finite CW complexes. But for general CW complexes a different definition is needed. The simplest thing to do is to associate to each spectrum K an Ω -spectrum K' and let $h^n(X) = \langle X, K'_n \rangle$. We obtain K' from K by setting $K'_n = \varinjlim \Omega^i K_{n+i}$, the mapping telescope of the sequence $K_n \rightarrow \Omega K_{n+1} \rightarrow \Omega^2 K_{n+2} \rightarrow \cdots$. The Ω -spectrum structure is given by equivalences

$$K'_n = \varinjlim \Omega^i K_{n+i} \simeq \varinjlim \Omega^{i+1} K_{n+i+1} \xrightarrow{\kappa} \Omega \varinjlim \Omega^i K_{n+i+1} = \Omega K'_{n+1}$$

The first homotopy equivalence comes from deleting the first term of the sequence $K_n \rightarrow \Omega K_{n+1} \rightarrow \Omega^2 K_{n+2} \rightarrow \cdots$, which has negligible effect on the mapping telescope. The next map κ is a special case of the natural weak equivalence $\varinjlim \Omega Z_n \rightarrow \Omega \varinjlim Z_n$ that holds for any sequence $Z_1 \rightarrow Z_2 \rightarrow \cdots$. Strictly speaking, we should let K'_n be a CW approximation to the mapping telescope $\varinjlim \Omega^i K_{n+i}$ in order to obtain a spectrum consisting of CW complexes, in accordance with our definition of a spectrum.

In case one starts with a suspension spectrum $K_n = \Sigma^n K$ it is not necessary to take mapping telescopes since one can just set $K'_n = \bigcup_i \Omega^i \Sigma^{i+n} K = \bigcup_i \Omega^i \Sigma^i K_n$, the union with respect to the natural inclusions $\Omega^i \Sigma^i K_n \subset \Omega^{i+1} \Sigma^{i+1} K_n$. The union $\bigcup_i \Omega^i \Sigma^i X$ is usually abbreviated to $\Omega^\infty \Sigma^\infty X$. Another common notation for this union is QX . Thus $\pi_i(QX) = \pi_i^\infty(X)$, so Q is a functor converting stable homotopy groups into ordinary homotopy groups.

It follows routinely from the definitions that the homology theory defined by a spectrum is the same as the homology theory defined by the associated Ω -spectrum. One may ask whether every homology theory is defined by a spectrum, as we showed for cohomology. The answer is yes if one replaces the wedge axiom by a stronger **direct limit axiom**: $h_i(X) = \varinjlim h_i(X_\alpha)$, the direct limit over the finite subcomplexes X_α of X . The homology theory defined by a spectrum satisfies this axiom, and the converse is proved in [Adams 1971].

Spectra have become the preferred language for describing many stable phenomena in algebraic topology. The increased flexibility of spectra is not without its price, however, since a number of concepts that are elementary for spaces become quite a bit more subtle for spectra, such as the proper definition of a map between spectra, or the smash product of two spectra. For the reader who wants to learn more about this language a good starting point is [Adams 1974].

Exercises

1. Assuming the first two axioms for a homology theory on the CW category, show that the direct limit axiom implies the wedge sum axiom. Show that the converse also holds for countable CW complexes.
2. For CW complexes X and Y consider the suspension sequence

$$\langle X, Y \rangle \xrightarrow{\Sigma} \langle \Sigma X, \Sigma Y \rangle \xrightarrow{\Sigma} \langle \Sigma^2 X, \Sigma^2 Y \rangle \longrightarrow \cdots$$

Show that if X is a finite complex, these maps eventually become isomorphisms. [Use induction on the number of cells of X and the five-lemma.]

3. Show that for any sequence $Z_1 \rightarrow Z_2 \rightarrow \cdots$, the natural map $\varinjlim \Omega Z_n \rightarrow \Omega \varinjlim Z_n$ is a weak homotopy equivalence, where the direct limits mean mapping telescopes.

4.G Gluing Constructions

It is a common practice in algebraic topology to glue spaces together to form more complicated spaces. In this section we describe two general procedures for making such constructions. The first is fairly straightforward but also rather rigid, lacking some homotopy invariance properties an algebraic topologist would like to see. The second type of gluing construction avoids these drawbacks by systematic use of mapping cylinders. We have already seen many special cases of both types of constructions, and having a general framework covering all these special cases should provide some conceptual clarity.

A **diagram of spaces** consists of an oriented graph Γ with a space X_v for each vertex v of Γ and a map $f_e: X_v \rightarrow X_w$ for each edge e of Γ from a vertex v to a vertex w , the words ‘from’ and ‘to’ referring to the given orientation of e . Commutativity of the diagram is not assumed. Denoting such a diagram of spaces simply by X , we define a space $\sqcup X$ to be the quotient of the disjoint union of all the spaces X_v associated to vertices of Γ under the identifications $x \sim f_e(x)$ for all maps f_e associated to edges of Γ . To give a name to this construction, let us call $\sqcup X$ the *amalgamation* of the diagram X . Here are some examples:

- If the diagram of spaces has the simple form $X_0 \xleftarrow{f} A \hookrightarrow X_1$ then $\sqcup X$ is the space $X_0 \sqcup_f X_1$ obtained from X_0 by attaching X_1 along A via f .
- A sequence of inclusions $X_1 \hookrightarrow X_2 \hookrightarrow \cdots$ determines a diagram of spaces X for which $\sqcup X$ is $\bigcup_i X_i$ with the weak topology. This holds more generally when the spaces X_i are indexed by any directed set.
- From a cover $\mathcal{U} = \{X_i\}$ of a space X by subspaces X_i we can form a diagram of spaces $X_{\mathcal{U}}$ whose vertices are the nonempty finite intersections $X_{i_1} \cap \cdots \cap X_{i_n}$ with distinct indices i_j , and whose edges are the various inclusions obtained by omitting some of the subspaces in such an intersection, for example the inclusions $X_i \cap X_j \hookrightarrow X_i$. Then $\sqcup X_{\mathcal{U}}$ equals X as a set, though possibly with a different topology. If the cover is an open cover, or if X is a CW complex and the X_i ’s are subcomplexes, then the topology will be the original topology on X .
- An action of a group G on a space X determines a diagram of spaces X_G , with X itself as the only space and with maps the homeomorphisms $g: X \rightarrow X$, $g \in G$, given by the action. In this case $\sqcup X_G$ is the orbit space X/G .